

Transparency, Not Alpha: Locating the Value of Islamic Finance in Ethics, Transparency, and Market Depth, with an Open Toolkit for Compliant Investing

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Abstract

This study asks whether Shariah-compliant asset classes offer exposure that conventional risk factors do not already price. Using daily data on listed Shariah and conventional funds from 1999 to 2026, drawn entirely from free secondary sources, it evaluates a pre-registered set of return, tail, systemic-risk, and connectedness hypotheses, and subjects the central result to an illiquidity-robustness battery. Three findings emerge. First, Shariah equity returns are not statistically distinguishable from a matched conventional benchmark that excludes financials and tilts toward quality and low leverage; the screening-related alpha is indistinguishable from zero in a well-powered sample, so negative screening merely relabels conventional factor exposures. Second, sukuk appears distinct at daily frequency, but that appearance does not survive scrutiny: at tradeable horizons that remove stale pricing the instrument is largely spanned by conventional fixed-income factors, and a paired bootstrap cannot reject equality with conventional securitized debt, the monthly R^2 gap to commercial mortgage-backed securities being 0.06 with a 95 percent interval of $[-0.05, 0.16]$. The daily distinctiveness reflects the instrument's severe illiquidity, or the slow repricing of its underlying real assets, which a single vehicle cannot separate. The broader lesson is methodological: distinctiveness attributed to Islamic asset classes can be an artifact of testing thinly traded instruments. Third, the screened-equity exposure is closely reproducible, so a compliant investor need not pay an active premium for it, though the passive-active fee gap involved is the general one and not an Islamic-specific cost. The contribution is a compliance-channel diagnostic and an illiquidity-robust corrective to a literature inclined to read distinctiveness into thin instruments, not a new performance claim. Read constructively, the results relocate rather than diminish the value of compliance, placing it in ethical commitment, in transparency, and in the priority of deepening Islamic capital markets, rather than in any measured return or risk distinction, and they yield an open, reproducible toolkit for building compliant exposures transparently.

Keywords: Islamic finance; Shariah screening; sukuk; illiquidity; systemic risk; maqasid al-shariah; reproducibility. **JEL:** G11, G12, G23, Z12.

1 Introduction

Islamic finance has grown into a multi-trillion-dollar industry, and sukuk, its asset-backed fixed-income instrument, is among its fastest-expanding segments and a central vehicle for real-economy and sovereign financing across the Gulf and the wider Muslim world. Whether the instruments produced by Shariah compliance are economically distinct from their conventional counterparts, rather than ethically distinct alone, is therefore a question of consequence for investors who buy them for diversification, for issuers who price them, and for regulators who

supervise the markets in which they trade. This paper asks that question and answers it carefully.

Islamic investing is commonly presented as offering both ethical compliance and a distinct, crisis-resistant risk-return profile. This paper shows that, when the comparison is constructed carefully and subjected to an illiquidity control, the distinctiveness does not survive, and draws out what does. The exercise is organized around a diagnostic that separates the two channels through which compliance operates and asks whether either leaves exposure conventional factors do not already price.

Negative screening excludes impermissible sectors, notably conventional financial intermediaries, and imposes leverage and income filters. *Positive structural requirement* insists that an instrument embody a claim on real productive assets and a sharing of risk, as in sukuk. A natural conjecture, and the one with which this study began, is that screening merely rearranges conventional exposures while asset-backing creates something new. The first half of the conjecture holds: screened equity is spanned by a matched conventional benchmark (Hoepner, Rammal, & Rezec, 2011; Walkshäusl & Lobe, 2012). The second does not. Sukuk looks distinct in daily data, but the appearance is produced by the instrument’s severe illiquidity; once stale pricing is removed by sampling at lower frequency, sukuk too is largely explained by conventional fixed-income factors, corroborating Almaskati (2023), who reaches the same conclusion from spread-based liquidity measures. The conclusion relocates rather than diminishes the value of compliance. Deflationary for the narrow asset-pricing question, it is constructive for the field, since the salient fact about the sukuk market is its illiquidity, a development and market-structure problem whose resolution would strengthen the very real-economy financing that compliance is meant to serve.

These observations define the problem and the gap. The distinctiveness ascribed to Islamic assets has been assessed piecemeal: Islamic equity against broad rather than matched benchmarks (Hoepner et al., 2011; Walkshäusl & Lobe, 2012), and sukuk through single methodologies (Almaskati, 2023; Najeeb, Bacha, & Masih, 2017), with the two channels of compliance never examined within one design. The research question is therefore singular: through which channel, if either, does Shariah compliance create exposure that conventional factors do not already price? Three objectives answer it, and they are also the paper’s contributions; the central result among them is a corroboration rather than a discovery.

The first objective, and contribution, is a reusable compliance-channel diagnostic that asks, for any compliant instrument, whether negative screening or positive structure leaves exposure conventional factors do not already span. The second is a multi-method corroboration and extension of Almaskati (2023): two distinct diagnostics converge on the illiquidity reading, a frequency-and-scale decomposition of comovement (a time-domain frequency ladder and a scale-domain wavelet, two views of the same frequency dependence) and an asset-backed control, supported by the Amihud illiquidity mechanism and a paired bootstrap for inference. The asset-backed control, which distinguishes a sukuk-specific effect from one general to illiquid securitized debt, is the genuine increment over the prior spread-based evidence (Table 8), obtained however on a single exchange-traded vehicle rather than a cross-section of issuances. The third is a transparency result: the fee gap between transparent and opaque compliant products is quantified within the compliant universe, so that the comparison recommends nothing impermissible. No excess return is claimed, and no distinct asset class is asserted.

2 Related literature

This study connects five strands and is, in its central result, a corroboration and extension rather than a discovery. The first strand assesses Shariah-screened equity. The recurrent finding is that Islamic equity does not deliver risk-adjusted outperformance once style is controlled, its behaviour reflecting tilts toward growth, quality, and low leverage (Hoepner et al., 2011;

Walkshäusl & Lobe, 2012); Elnahas, Ismail, El-Khatib, and Hassan (2021) likewise report that Shariah-labelled firms are neither less risky nor less failure-prone than their peers. The present paper sharpens this by spanning Islamic returns against a *matched* ex-financials, quality, and low-beta benchmark, using the five-factor model with momentum (Fama & French, 2015), so that the residual is the Shariah-specific increment rather than a comparison with the broad market.

The second strand concerns the distinctiveness of sukuk, and it is more developed than is sometimes acknowledged. Studies using multi-scale wavelet correlation and coherence (Haque, Chowdhury, Buriev, Bacha, & Masih, 2018; Najeeb et al., 2017) and frequency-domain connectedness (Billah, Kapar, Hassan, Pezzo, & Rabbani, 2024) document that sukuk co-move less with conventional bonds at high frequency than at low, and typically read this as a diversification opportunity. Most directly, Almaskati (2023), using spread-based liquidity measures and propensity matching, shows that sukuk are significantly less liquid than matched conventional bonds and argues that the apparent diversification benefit is *driven by illiquidity rather than underlying economic factors*; sukuk illiquidity is independently recognised as a structural problem (Ferdian & Dewi, 2017). Our contribution to this strand is therefore not the claim but its multi-method corroboration: a frequency-and-scale decomposition of comovement (a frequency ladder of factor regressions and a wavelet decomposition that view the same frequency dependence from the time and scale domains) and an asset-backed control, with the Amihud measure supplying the mechanism and a paired bootstrap the inference that bounds the residual.

The third strand, largely outside the Islamic-finance literature, supplies the mechanism. Stale and infrequent pricing smooths returns, depresses measured co-movement, and inflates apparent idiosyncrasy, formalised for return smoothing by Getmansky, Lo, and Makarov (2004) and for thin-trading beta bias by Dimson (1979), with illiquidity measured following Amihud (2002). Importing this lens is what links the diversification evidence of the second strand to its likely cause.

The fourth strand measures systemic and connectedness risk with conditional value-at-risk (Adrian & Brunnermeier, 2016), marginal expected shortfall (Acharya, Pedersen, Philippon, & Richardson, 2017), and generalised variance-decomposition connectedness (Diebold & Yilmaz, 2012; Pesaran & Shin, 1998), complemented by regime identification (Hamilton, 1989). These tools have already been applied to sukuk, generally finding low connectedness and limited systemic risk (Al-Yahyaee, Shahzad, Mensi, & Yoon, 2021; Billah et al., 2024). We re-read such findings through the illiquidity lens, since the low connectedness of a stale-priced instrument is in part mechanical.

The fifth strand is normative. The maqasid literature locates the value of Islamic finance in real-economy linkage and risk-sharing (Choudhury, 2018; El-Gamal, 2006; Furqani & Haneef, 2023) and calls for reform toward substance over form (Hassan, 2020). We connect this to the empirics through a compliance-channel diagnostic that asks whether negative screening or positive structure creates exposure conventional factors do not span. The paper’s contribution is accordingly one of corroboration, identification, and synthesis. It corroborates Almaskati (2023), whose illiquidity conclusion it reaches by independent methods, and adds two elements the spread-based design omits: an asset-backed control that asks whether the effect is sukuk-specific or general to illiquid securitized debt, and an embedding within a compliance-channel diagnostic alongside a matched-benchmark equity null. Because the corroboration rests on a single exchange-traded vehicle rather than a cross-section of issuances, it adds method and angle rather than evidentiary weight, and it leaves open whether some sukuk are genuinely distinct once liquidity is controlled.

Table 1: The maqasid crosswalk: principle, channel, prediction, verdict.

Principle	Channel	Predicted market signature	Verdict in this study
Prohibition of riba; exclusion of financials	Negative screening	Lower left-tail co-movement with financials	Modest: tail dependence 0.55 vs 0.63
Leverage and debt limits	Negative screening	Quality and low-investment tilt	Holds, but fully spanned by a matched benchmark
Real-asset backing	Positive structure	A return process not spanned by conventional factors	Not supported once illiquidity is controlled (Table 7)
Risk-sharing	Positive structure	Insulation from system shocks	Apparent insulation coincides with stale-price illiquidity
Prohibition of gharar and maysir	Negative screening	Lower portfolio tail risk	Not isolated in screened equity

3 A maqasid-based diagnostic

Islamic finance is grounded in the objectives of the Shariah, among them the prohibition of interest, the avoidance of excessive uncertainty and speculation, the requirement that financial claims rest on real assets, and the preference for risk-sharing over risk-transfer (El-Gamal, 2006). Rather than ask whether compliant portfolios outperform, this study asks through which channel compliance operates and whether that channel creates exposure beyond conventional pricing. Table 1 sets out the predictions and the verdict.

The framework’s value is diagnostic, and the verdict it returns is deflationary. Negative screening relabels conventional factor exposures, so screened equity carries no content beyond the market and a quality tilt. Positive structure appears to create something new only at daily frequency; that appearance is the mechanical signature of infrequent pricing, whether from illiquidity or from the slow repricing of real underlying assets, which a single vehicle cannot separate, and it largely dissolves under an illiquidity control. The pre-registered tail predictions are read in this light. Several candidate signatures of structural distinctiveness, an unspanned residual, weak connectedness, and apparent insulation, are also exactly what infrequent pricing produces, so the surviving differences are small and must be interpreted with care. The contribution of the framework is thus not the confirmation of a distinct Islamic asset class, which the data refuse, but a disciplined way of asking the question and an honest accounting of the answer.

4 Data

All series are obtained without charge. Daily total-return series for listed funds are from public price providers; Fama-French factors are from the Ken French Data Library; macroeconomic and credit series are from the Federal Reserve Economic Database. The sample runs from January 1999 to June 2026, subject to instrument inception. Shariah equity is proxied by tradable funds (SPUS, HLAL, UMMA) and long-run Shariah mutual funds (AMANX from 1999, AMAGX, IMANX); sukuk by the global sukuk fund SPSK; conventional benchmarks and style, sector, fixed-income, and asset-backed factors by the corresponding liquid funds. Using tradable vehicles forecloses survivorship and index-construction concerns, at the cost of expense drag and, for sukuk, material illiquidity, measured below and central to the analysis. The complete instrument set is listed in Table 2; the pipeline is reproducible from free data with a single command. Several of the listed series, notably the GCC equity, oil, policy-uncertainty, and gold series, were pulled for context and the asynchronous cross-market checks but do not enter the reported specifications, which rest on the US-listed instruments, so that the data listed

and the data analysed are not conflated.

Table 2: Instrument set (exchange tickers).

Group	Tickers
Shariah equity (ETF)	SPUS, HLAL, UMMA
Shariah equity (funds, long)	AMANX, AMAGX, IMANX
Sukuk	SPSK
Conventional fixed income	AGG, LQD, HYG, EMB, TLT, IEF, BGRN
Conventional asset-backed	MBB, VMBS, CMBS
Market / benchmark	SPY, ACWI, URTH, VFINX, ^GSPC
Style / sector factors	QUAL, MTUM, USMV, VLUE, XLF, QQQ, XLK, SMH, MAGS
International Islamic / EM	ISDU.L, ISDE.L, EEM, EFA
GCC / macro / gold	QAT, KSA, DX-Y.NYB, PAXG-USD
FRED series	DGS2, DGS10, VIXCLS, NFCL, BAMLH0A0HYM2, DFF, DCOILBRENTU, USREC

5 Methods and pre-registration

To guard against specification search, the estimands, comparisons, and directional predictions were fixed before estimation. Returns enter as simple returns to match the Fama-French factors; standard errors are Newey-West (Newey & West, 1987). The matched benchmark removes financials from the market and reweights, isolating any Shariah-specific increment beyond a no-financials, leverage-capped rule. Hypotheses on equity: H0, Islamic equity does not differ from the matched benchmark; H1, Islamic equity loads on quality and low-investment factors; H2, out-of-sample spanning alpha is zero. Six tail, systemic, and connectedness predictions were pre-registered from *maqasid*: P1, lower-tail dependence with financials below the market’s; P2, a near-zero sukuk marginal expected shortfall; P3, sukuk lower-tail decoupling from equity; P4, compression of the downside bank-beta of Islamic equity; P5, an Islamic-equity systemic contribution no greater than financials’; and P6, sukuk as the net receiver in a connectedness system with Islamic equity behaving as the market. The sukuk illiquidity result is exploratory, not pre-registered, and is labelled E. The central sukuk result is then subjected to an illiquidity battery: the factor decomposition is re-run across daily, weekly, and monthly horizons and with Dimson lags (Dimson, 1979), and illiquidity is measured by the Amihud ratio (Amihud, 2002). That stale pricing inflates apparent idiosyncrasy in thinly traded assets is a known caution (Getmansky et al., 2004). Methods comprise factor attribution (Fama & French, 2015), nonparametric copula tail dependence, conditional value-at-risk (Adrian & Brunnermeier, 2016) and marginal expected shortfall (Acharya et al., 2017), dynamic conditional correlation (Engle, 2002), generalized connectedness (Diebold & Yilmaz, 2012; Pesaran & Shin, 1998), a Markov-switching regime model (Hamilton, 1989), and a wavelet scale-decomposition of comovement by frequency as an independent check on the illiquidity reading.

5.1 Key definitions

Excess returns are regressed on the factor set, $r_{i,t} - r_{f,t} = \alpha_i + \beta_i' \mathbf{F}_t + \varepsilon_{i,t}$, and the matched-benchmark spanning test replaces $\mathbf{F}_t$ with the ex-financials market, a quality factor, and a low-beta factor; $H0$ is $\alpha_i = 0$.

Generalised forecast-error-variance connectedness uses, for an estimated VAR with moving-average matrices A_h and residual covariance Σ ,

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (\mathbf{e}_i' A_h \Sigma \mathbf{e}_j)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i' A_h \Sigma A_h' \mathbf{e}_i}, \quad \tilde{\theta}_{ij} = \frac{\theta_{ij}}{\sum_k \theta_{ik}}, \quad C = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}, \quad (1)$$

where C is the total connectedness index and net directional connectedness is $\sum_i \tilde{\theta}_{ki} - \sum_i \tilde{\theta}_{ik}$.

Systemic exposure is measured by ΔCoVaR and marginal expected shortfall. From a quantile regression of the system X^{sys} on asset X_i at level q , $\Delta\text{CoVaR}_q^i = \hat{\beta}_q(\text{VaR}_q(X_i) - \text{VaR}_{0.5}(X_i))$, and $\text{MES}_i = \mathbb{E}[r_i \mid r_m \leq \text{VaR}_\alpha(r_m)]$ with $\alpha = 5\%$. Lower-tail dependence is estimated nonparametrically as $\lambda_L = \widehat{\Pr}(U_i \leq q \mid U_j \leq q)$ at $q = 0.05$ on the empirical copula. Dynamic hedge ratios use the DCC correlation ρ_t and conditional volatilities, $\beta_t = \rho_t h_{i,t}/h_{j,t}$, with hedging effectiveness $1 - \text{Var}(r_i - \beta_t r_j)/\text{Var}(r_i)$.

5.2 Construction and parameters

Returns are simple (`pct_change`) computed on each series' own trading calendar, with daily moves above $|0.8|$ masked as data errors; weekly returns use Friday closes and monthly returns month-end closes. The matched benchmark is $\text{ex_fin}_t = (\text{SPY}_t - 0.13 \text{XLF}_t)/0.87$, with 0.13 the approximate S&P 500 financials weight. Constructed factors: quality = $\text{QUAL} - \text{SPY}$, low-beta = $\text{USMV} - \text{SPY}$; fixed-income factors duration = IEF , IG credit = $\text{LQD} - \text{IEF}$, high yield = $\text{HYG} - \text{LQD}$, emerging = $\text{EMB} - \text{LQD}$, green = $\text{BGRN} - \text{AGG}$, asset-backed = $\text{MBB} - \text{IEF}$. Estimation settings are collected in Table 3.

Table 3: Estimation parameters.

Component	Setting
Standard errors	Newey-West HAC, 5 lags
Out-of-sample split	train $\leq 2019-12-31$; test 2020–2026
Connectedness	VAR(2), generalised FEVD horizon $H = 10$; system $\{\text{SPUS}, \text{SPSK}, \text{SPY}, \text{XLF}, \text{EMB}, \text{PAXG}\}$
DCC	univariate GJR-GARCH(1,1,1)- t ; DCC(1,1) by QMLE
Tail dependence	nonparametric empirical copula, $q = 0.05$
CoVaR / quantile β	quantile regression, $\tau = 0.05$; $\tau \in \{0.05, 0.25, 0.5, 0.75, 0.95\}$
MES	mean return on worst-5% SPY days
Illiquidity battery	horizons D / W-FRI / month-end; Dimson 1 lag (weekly)
Wavelet	stationary wavelet transform, db4, 6 levels
Bootstrap (R^2 gap)	paired, 2,000 draws, monthly
Markov-switching	2 regimes, switching variance, on SPY
Amihud illiquidity	$ r_t /(\text{price}_t \cdot \text{volume}_t)$

Event-study windows: Lehman (2008-09-08 to 2008-11-21), COVID (2020-02-19 to 2020-03-23), SVB (2023-03-08 to 2023-03-24), and Credit Suisse (2023-03-15 to 2023-03-31) for the credit leg; taper tantrum (2013-05-22 to 2013-06-24), 2018-Q4 (2018-10-01 to 2018-12-24), and the 2022 hiking cycle (2022-01-01 to 2022-10-31) for the rate leg; ChatGPT (2022-11-30 to 2023-02-28) and Nvidia (2023-05-24 to 2023-07-31) for the technology leg.

6 Results

6.1 Descriptive statistics and diagnostics

Table 4 reports the moments and pre-estimation diagnostics for the principal daily return series. All series are stationary: the augmented Dickey-Fuller test rejects a unit root at the one percent level for every series, and the KPSS statistic is correspondingly small. Returns are markedly non-normal, with excess kurtosis throughout (Jarque-Bera rejects normality at one percent for every series), which motivates the heteroskedasticity- and autocorrelation-consistent standard errors used below. Ljung-Box tests reject the absence of autocorrelation, supporting the vector-autoregressive treatment, and Engle's LM test finds pronounced ARCH effects in every series ($p < 0.01$), which justifies the GARCH and dynamic conditional correlation specifications. Sukuk is the exception: its modest excess kurtosis (3.2, against 8 to 50 for the other

Table 4: Descriptive statistics and unit-root tests for daily returns (%). Jarque-Bera, Ljung-Box(10), and ARCH(10) reject their nulls at 1% for all series.

	Mean	SD	Skew	Ex. kurt	ADF	KPSS
SPUS	0.076	1.34	-0.11	8.0	-12.9	0.04
HLAL	0.070	1.28	-0.37	10.5	-13.0	0.04
AMANX	0.039	0.97	-0.15	8.8	-17.2	0.14
SPSK	0.004	0.34	-0.01	3.2	-9.5	0.15
SPY	0.040	1.21	+0.04	11.7	-20.3	0.33
XLF	0.039	1.78	+0.31	14.9	-15.0	0.10
AGG	0.013	0.33	-1.79	49.5	-16.0	0.22
EMB	0.021	0.68	-1.60	35.7	-14.9	0.07
MBB	0.011	0.29	-0.03	11.8	-17.6	0.29
CMBS	0.010	0.35	-1.70	45.6	-16.8	0.09

ADF: augmented Dickey-Fuller statistic (all $p < 0.01$, rejecting a unit root). KPSS: stationarity statistic (5% critical value 0.463; not rejected). Sample is each series' full daily history within 1999–2026.

Table 5: Factor loadings (FF5+MOM, HAC t -statistics) and matched-benchmark spanning.

	Mkt-RF	HML	RMW	CMA	α (ann.%)	spanning α t
SPUS	0.98	-0.23	+0.12	+0.07	+2.5 (1.7)	1.1
HLAL	0.96	-0.09	+0.09	+0.08	+1.5 (1.0)	0.8
AMANX	0.77	-0.02	+0.24	+0.27	-0.5 (-0.5)	-0.4
AMAGX	0.89	-0.20	-0.06	-0.03	+3.9 (3.4)	1.3

Spanning alpha is on a matched ex-financials, quality, and low-beta benchmark; all insignificant. AMAGX's positive raw alpha reflects active management and is treated as a caveat.

series) is itself consistent with the return smoothing that infrequent pricing induces (Getmansky et al., 2004), foreshadowing the illiquidity reading of Section 6.4. For the connectedness system the Hannan-Quinn criterion selects a second-order VAR, which is adopted; the Akaike and final-prediction-error criteria select higher orders and the Schwarz criterion a first-order model.

The results are reported in two parts. The pre-registered hypotheses are examined first: the equity spanning tests and the tail and systemic predictions derived in Section 3. An exploratory analysis of sukuk's apparent distinctiveness, prompted by the rejection of several of those predictions, follows and is labelled as such.

6.2 Screened equity is a transparent factor portfolio

Islamic equity loads significantly on quality and conservative-investment factors and tilts to growth and large capitalization (Table 5). Once these tilts are controlled by a matched ex-financials quality benchmark, the spanning alpha is statistically indistinguishable from zero for every Shariah series, with t -statistics below 1.4 in samples as large as 6,872 daily observations, and it remains insignificant out of sample. The screening-related advantage is exposure, not skill, and is reproducible by conventional means. The null is insensitive to the matched benchmark's financials weight: the spanning alpha is 1.4 to 1.5 percent per year with a t -statistic near 1.1 for weights from 0.10 to 0.16. The one significant raw alpha, AMAGX at 3.9 percent ($t = 3.4$), belongs to an actively managed growth fund rather than to screening, and it reverses out of sample (annualized -1.2 percent, $t = -0.4$); the passive screening proxies carry no such alpha.

6.3 Tails and systemic risk: the pre-registered predictions

On the worst five percent of market days, screened equity is not protective; its mean return is more negative than the market's and its system-tail contribution exceeds that of financials

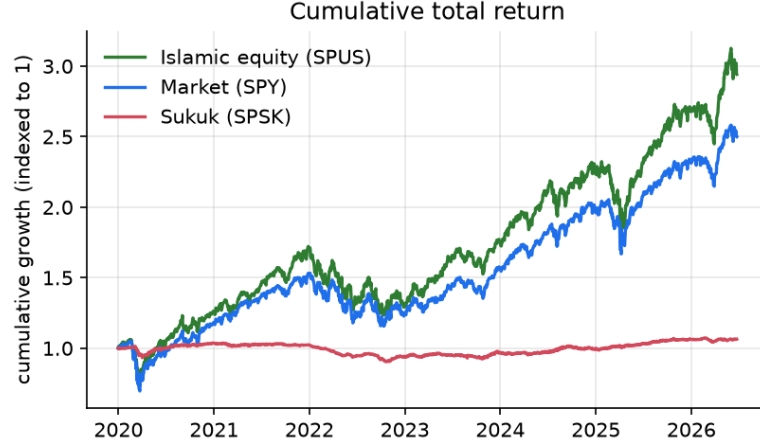


Figure 1: Cumulative total return of Shariah equity, the market, and sukuk. Islamic equity tracks the market closely; sukuk is a low-volatility series whose apparent distinctiveness is examined in Section 6.4.

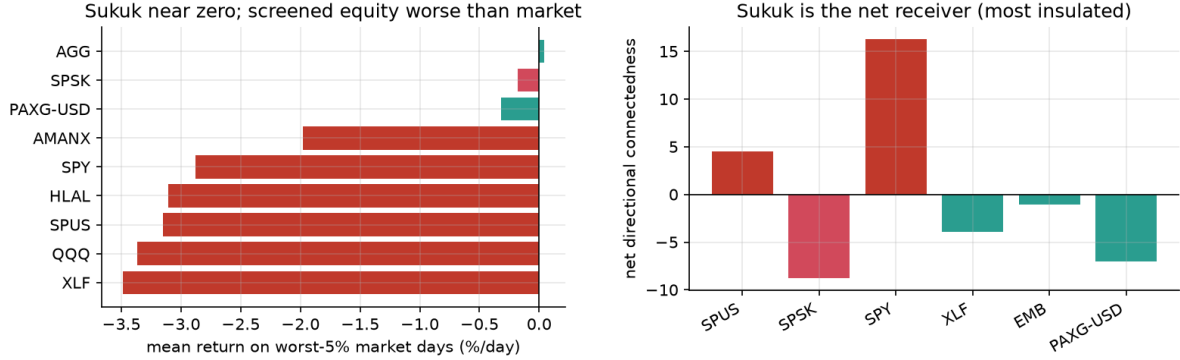


Figure 2: Left: marginal expected shortfall on the worst-5% market days; sukuk is near zero while screened equity is worse than the market. Right: net directional connectedness; sukuk is the net receiver, and screened equity behaves like the market.

(Table 6). The pre-registered prediction that the financials exclusion lowers tail co-movement with banks holds (lower-tail dependence 0.55 against the market's 0.63), but the predictions that screened equity carries lower systemic risk than financials, and that its downside bank-beta is compressed, are both rejected. The former rejection does not imply that Islamic equity is a distinct source of systemic risk: it contributes to the system tail because it is, statistically, the market, its received connectedness of 64.8 percent matching the market's 68.0 percent. This is consistent with Al-Yahyaee et al. (2021), who find no *distinct* systemic risk in Islamic equity or sukuk. Sukuk records a near-zero loss and the smallest systemic contribution among the risk assets, and is the most insulated node in a generalized connectedness system (total spillover 48.7 percent; sukuk net directional -8.8 , received spillover 28 percent). These last readings are also exactly what an illiquid, stale-priced instrument would produce, a possibility the exploratory analysis of Section 6.4 investigates directly; a conventional aggregate bond fund is in any case marginally more resilient on the same worst-day metric. A Markov-switching model dates a crisis regime at 28 percent of days and corroborates the equity reading: screened equity falls with the market in the crisis state.

Table 6: Tail and systemic risk. MES is the mean return on the worst-5% market days.

	λ_L with financials	MES (%/day)	ΔCoVaR (%/day)
SPUS (Islamic equity)	0.55	−3.15	−2.03
SPY (market)	0.63	−2.88	—
XLF (financials)	1.00	−3.49	−1.35
SPSK (sukuk)	0.15	−0.18	−0.59
AGG (conv. bonds)	—	+0.04	—

The low sukuk figures coincide with its illiquidity (Table 7) and are interpreted accordingly. ΔCoVaR is undefined for the market (SPY), which is the reference system, and is not computed for AGG; a dash denotes not applicable.

6.4 Exploratory: sukuk’s apparent distinctiveness is illiquidity

The rejection of the pre-registered safe-haven and bank-beta predictions raises a question the pre-registration did not anticipate: is the apparent distinctiveness of sukuk a real property, or an artifact of how the instrument is priced? The analysis that follows is exploratory and is labelled as such. Its conclusion corroborates Almaskati (2023), who reaches it with spread-based liquidity measures on a cross-section of issuances; the contribution here is the convergence of two distinct diagnostics, a frequency-and-scale decomposition of comovement and an asset-backed control absent from that design, supported by the Amihud mechanism and a paired bootstrap, on the same answer, albeit on a single exchange-traded vehicle. A conventional factor decomposition explains only a quarter of daily sukuk variation, which at first reading suggests a distinct asset class. It does not survive an illiquidity control (Table 7). As the sampling horizon lengthens from daily to weekly to monthly, the explained share rises from 0.25 to 0.59 to 0.81, converging toward the values for conventional securitized debt (0.86 for commercial mortgage-backed securities, 0.88 for agency mortgage-backed securities). Adding Dimson lags at weekly frequency raises the explained share by a further six points. A paired bootstrap cannot reject equality with conventional securitized debt: the monthly R^2 gap to commercial mortgage-backed securities is 0.06 with a 95 percent interval of $[-0.05, 0.16]$. The interval is wide: with ninety-one monthly observations it cannot reject equality, but neither can it reject a gap as large as 0.16, so it bounds rather than establishes non-distinctiveness. The instrument’s Amihud illiquidity is ten to one hundred fifty times that of conventional bond funds, which supplies a mechanism, since infrequent prices do not co-move and so appear idiosyncratic. Whether the small remaining gap reflects residual illiquidity or the slow repricing of real underlying assets cannot be resolved with a single vehicle. What can be said is that the daily distinctiveness is, on the available evidence, an artifact of infrequent pricing rather than a return process beyond conventional finance.

A frequency-domain analysis confirms the mechanism from a different direction. Decomposing returns with a stationary wavelet transform and correlating the detail coefficients by timescale (Figure 4), the comovement of sukuk with conventional bonds rises from about 0.34 at the two-day scale to between 0.75 and 0.90 at the two-to-three-month scale, whereas the comovement of screened equity with the market is near 0.96 at every scale. Sukuk’s apparent independence is confined to the high-frequency band where stale pricing dominates; at the low frequencies that reflect fundamentals it moves with conventional fixed income. Screened equity is the market at all frequencies. The frequency ladder and the wavelet approach the same frequency dependence from the time and scale domains respectively, and so deliver the same verdict by construction; the genuinely independent corroboration comes from the asset-backed control of Table 8, which uses different instruments entirely.

The asset-backed control isolates what is sukuk-specific from what is generic to illiquid securitized debt, and it is the increment this paper adds to the spread-based evidence (Table 8). Spanned at a common weekly horizon against the same conventional fixed-income factors, plain bond funds are almost fully explained (investment-grade and emerging-market credit R^2 near

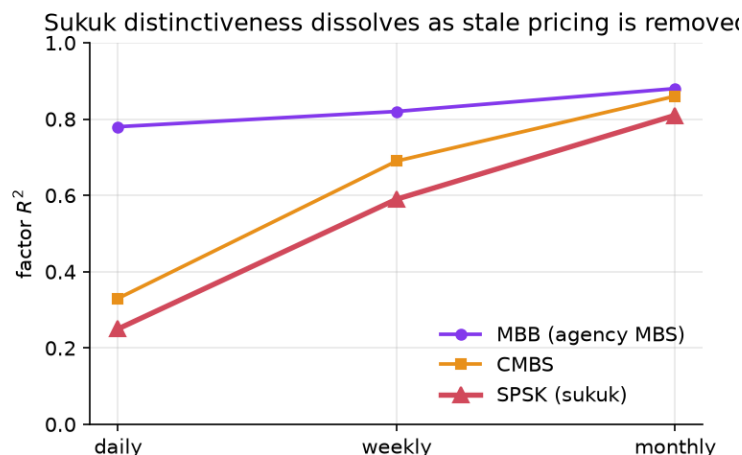


Figure 3: The headline result. As the sampling horizon lengthens and stale pricing is removed, sukuk’s factor R^2 rises toward that of conventional securitized debt; the daily “distinctiveness” is largely illiquidity.

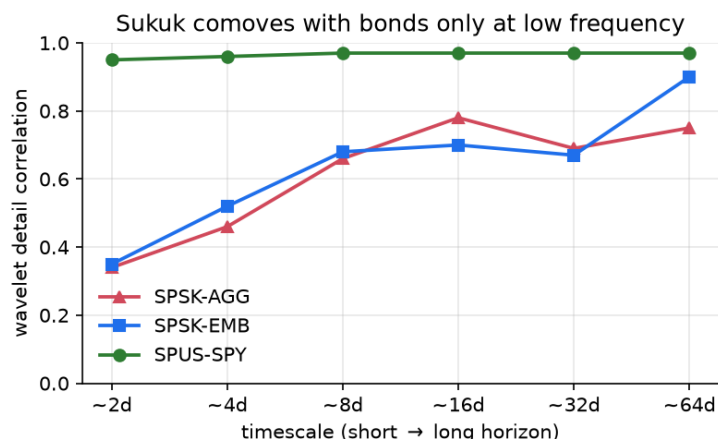


Figure 4: Wavelet detail-coefficient correlation by timescale. Sukuk-bond comovement rises with horizon, so the high-frequency decoupling is stale pricing; equity-market comovement is scale-invariant.

unity), agency and commercial mortgage-backed securities partly so (0.82 and 0.69), and sukuk the least (0.59, 41 percent unexplained). Sukuk therefore sits at the illiquid end of a securitization spectrum rather than apart from it. Adding a conventional asset-backed (agency-MBS) factor to the sukuk regression, however, raises the explained share by only one point (0.59 to 0.60) despite a significant loading, so a small sukuk-specific component survives at weekly frequency that conventional asset-backed risk does not absorb, consistent with the residual the monthly bootstrap cannot rule out.

6.5 Transparency and the cost of opacity

The screened-equity exposure is closely reproducible from inexpensive public building blocks: a benchmark of the ex-financials market and a quality fund, estimated through 2019, tracks the Shariah fund out of sample with a correlation of 0.98 and an annualized tracking error of 4.4 percent. A rules-based balanced combination of a transparent equity sleeve and a sukuk sleeve, net of an assumed twenty-basis-point cost, broadly matches a conventional 60/40 portfolio on

Table 7: Illiquidity battery: factor R^2 by horizon. Distinctiveness dissolves as stale pricing is removed.

Horizon	SPSK (sukuk)	CMBS	MBB (agency MBS)
Daily	0.25	0.33	0.78
Weekly	0.59	0.69	0.82
Monthly	0.81	0.86	0.88

Dimson lag at weekly frequency: R^2 rises 0.59 to 0.65. Amihud illiquidity: SPSC ≈ 14.7 , versus 0.1 to 1.4 for conventional bond funds (scaled $\times 10^9$).

Table 8: Asset-backed control: weekly factor R^2 across the securitization spectrum. Sukuk is the least spanned; an added agency-MBS factor absorbs little of its residual.

Instrument	R^2	Unexplained (%)	α (ann.%)
LQD (IG credit)	1.00	0	0.0
EMB (EM sovereign)	1.00	0	0.0
AGG (aggregate bond)	0.98	2	0.4
VMBS (agency MBS)	0.83	17	0.2
MBB (agency MBS)	0.82	18	0.0
CMBS (commercial MBS)	0.69	31	1.8
SPSK (sukuk)	0.59	41	0.5

Weekly (W-FRI) returns on the conventional fixed-income factor set; HAC(4) t -statistics. Adding an agency-MBS factor (MBB-IEF) to the sukuk regression raises R^2 from 0.59 to 0.60 ($\Delta R^2 = +0.01$; MBS loading +0.18, $t = 2.6$), so conventional asset-backed risk leaves a small sukuk-specific residual.

a risk-adjusted basis (Sharpe 0.84 in sample and 0.66 in the post-2022 stress window, against 0.76 and 0.64) and reduces drawdown relative to holding Shariah equity alone, though the out-of-sample difference is within noise and is reported as a match rather than an improvement. To avoid recommending anything impermissible, the cost of opacity is measured within the compliant universe: transparent passive Shariah exchange-traded funds charge roughly 0.40 to 0.65 percent annually against 0.86 to 1.30 percent for active, opaque Shariah funds, a difference of about fifty-five to ninety basis points per year, or roughly one thousand to seventeen hundred dollars per ten thousand invested over ten years. Both sides are compliant. This gap is the general active-passive fee differential found throughout asset management rather than an Islamic-specific cost of opacity; its narrower relevance here is that, because the screened-equity exposure is reproducible, a compliant investor need not pay an active premium to obtain it. Tables 9 and 10 report the fee comparison and the balanced-portfolio back-test.

Table 9: Fees within the compliant universe and a ten-year cost on \$10,000 at 8% gross.

Compliant product	expense (%/yr)	10y cost (\$)
ISDU.L, SPUS, HLAL, UMMA (passive)	0.40–0.65	786–1,265
AMAGX, AMANX, IMANX (active)	0.86–1.30	1,659–2,462

6.6 Robustness and auxiliary results

Three further results are reported for completeness and all point the same way. First, the sukuk convergence is inferential, not eyeballed: a paired bootstrap on the monthly factor decomposition gives a sukuk-to-commercial-mortgage R^2 gap of 0.06 with a 95 percent interval of $[-0.05, 0.16]$, so equality cannot be rejected. Second, event studies around dated credit, rate, and technology shocks, measured against the matched benchmark, find no significant abnormal returns in any leg (Table 11); the visible crisis behaviour is the mechanical bank-exclusion effect, not a Shariah

Table 10: Balanced-portfolio back-test, net of 20 bps. Sharpe and maximum drawdown.

	Sharpe (full)	Sharpe (2022+)	maxDD (%)
70/30 Islamic equity + sukuk	0.84	0.66	−23.6
Islamic equity only (SPUS)	0.85	0.65	−30.8
60/40 conventional	0.76	0.64	−21.7

property. Third, the equity null travels: the spanning of Islamic equity by the market and a quality factor recurs across vehicles, subject to the asynchronous pricing of non-US-listed funds (Table 12).

Table 11: Event studies vs the matched benchmark: pooled abnormal returns by leg.

Leg	events	mean aligned CAR (%)	share correct sign	pooled t
Credit (Lehman, SVB, COVID)	24	+0.46	0.67	0.50
Rate (taper, 2018Q4, hiking)	15	+1.50	0.60	0.77
Technology (ChatGPT, Nvidia)	14	−0.93	0.43	−1.31

Table 12: Cross-market spanning of Islamic equity (HAC t -statistics).

	R^2	α (ann.%)	$t(\alpha)$
SPUS $\sim$ market + quality (US ETF)	0.92	+2.6	1.3
ISDU.L $\sim$ market + quality (US, listed abroad)	0.29	+4.3	1.6
ISDE.L $\sim$ EM market (EM Islamic)	0.35	+2.5	0.7

Low R^2 for the non-US-listed funds reflects asynchronous pricing, not distinct content; the spanning alpha is insignificant throughout.

7 Discussion

The results admit a coherent reading once each is placed against its hypothesis and the prior literature. Table 13 maps every pre-registered hypothesis to its result, status, and the consistent prior work; the paragraphs that follow interpret the pattern.

The equity findings confirm the pre-registered hypotheses H0 to H2 and align with a settled literature: Islamic equity is a style portfolio, and screening adds no risk-adjusted increment once a matched benchmark absorbs the tilts. This reproduces, with a sharper benchmark, the conclusions of Walkshäusl and Lobe (2012) and Hoepner et al. (2011), and is consistent with Elnahas et al. (2021), who find Shariah-labelled firms neither less risky nor less failure-prone than their peers.

The tail and systemic results are mixed against the pre-registered predictions, and the pattern is itself informative. The bank-decoupling prediction holds, but the predictions that screened equity is less systemic than financials and exhibits a compressed downside bank-beta are rejected. As discussed, this reflects screened equity being statistically the market rather than a distinct systemic entity, which reconciles with Al-Yahyaee et al. (2021), who report no *distinct* systemic risk in Islamic markets. The sukuk readings, a near-zero marginal expected shortfall and the most insulated connectedness position, confirm the connectedness hypothesis and align with Billah et al. (2024), but they coincide with the instrument’s illiquidity and are therefore read with caution rather than as evidence of a stabilising property.

It is the rejection of the sukuk safe-haven predictions that motivates the exploratory result, and there the paper corroborates Almaskati (2023): the apparent distinctiveness and diversification benefit of sukuk are, on the evidence, an artifact of illiquidity rather than a distinct return

Table 13: Summary of hypothesis tests: prediction, result, status, and prior literature.

	Prediction	Result	Status	Prior literature
H0	Islamic equity = matched benchmark ($\alpha = 0$)	spanning α insignificant	supported	Hoepner et al. (2011); Walkshäusl and Lobe (2012)
H1	loads on quality / low investment	RMW, CMA > 0	supported	Elnahas et al. (2021)
H2	out-of-sample $\alpha = 0$	insignificant	supported	—
P1	tail dependence with financials $<$ market	0.55 vs 0.63	supported	—
P2	sukuk MES ≈ 0	-0.18	supported	Al-Yahyaee et al. (2021)
P3	sukuk decouples from equity in the tail	$\lambda_L = 0.17$ (vs market), $\lambda_L > \lambda_U$	partial	Najeeb et al. (2017)
P4	downside bank-beta compressed	flat across quantiles	rejected	—
P5	systemic contribution $\leq$ financials	exceeds financials	rejected	Al-Yahyaee et al. (2021)
P6	sukuk net receiver; equity = market	NET -8.8 ; FROM $\approx$ market	supported	Billah et al. (2024)
E	sukuk distinctiveness is illiquidity (exploratory)	R^2 0.25 $\rightarrow$ 0.81, converges	supported	Almaskati (2023)

process. The frequency-dependence that Najeeb et al. (2017) and Haque et al. (2018) document and read as a diversification opportunity is re-interpreted here as the signature of stale pricing, and is confirmed by a frequency-and-scale decomposition of comovement and, independently, by the asset-backed control, with the Amihud mechanism and a paired bootstrap completing the picture. The asset-backed control (Table 8) is the genuine increment over the prior spread-based evidence, and it also marks the honest limit: a single vehicle cannot separate residual illiquidity from genuine sukuk-specific risk, so the question is not settled across the asset class.

Read through the compliance-channel diagnostic, the results say that neither negative screening nor, once illiquidity is controlled, the structure of the sukuk examined here yields exposure that conventional factors do not already price. The value of compliance, on this evidence, does not lie in a differentiated risk-return profile, which is the premise the next section develops.

8 From asset pricing to Islamic economics

The result is statistical, but its lesson for Islamic economics is substantive and, in a sense, encouraging. A discipline that grounds finance in real assets and shared risk need not rest its case on a claim of differentiated market returns, which the careful evidence does not support. What the evidence points to, on a single internationally listed vehicle, is the illiquidity and segmentation of the sukuk market, severe enough to masquerade as a distinct risk-return profile. Should this hold on Gulf-listed instruments, it would reframe the policy question, suggesting that the contribution of sukuk to the real economy is constrained less by any pricing property than by the thinness of its secondary market.

Three implications would then follow for the Gulf and for institutions such as the Qatar Financial Centre and the Islamic Financial Services Board. First, market deepening and liquidity provision, not product proliferation, are the priority, because illiquidity, rather than any pricing property, is the more plausible constraint on the very real-economy financing the maqasid favour. Second, a reproducible, free-data methodology lowers the cost and informational barrier for smaller Gulf issuers and managers to construct and audit compliant exposures, supporting a

wider market. Third, treating screening and structure as distinct channels gives supervisors a sharper question, namely how much of an institution’s compliance is exclusionary relabelling and how much is genuine asset-backed, risk-sharing finance. The empirics use internationally listed instruments and a single sukuk vehicle; the illiquidity finding in particular should be re-examined on Gulf-listed sukuk, where segmentation may differ.

9 The digital frontier

The conference theme concerns autonomous systems. The most defensible digital implication of these results is narrow. Since the binding problem is illiquidity, the promise of tokenization for Islamic finance lies in secondary-market liquidity and fractional access for asset-backed instruments, not in tokenizing screened equity, which carries no distinct content. The open, rules-based methodology developed here is a candidate substrate for auditable index construction and for a checkable compliance signal. Three difficulties are acknowledged: the oracle problem of verifying off-chain asset-backing, the Shariah governance of code, which encoding does not transfer, and liquidity, which the documented sukuk illiquidity would make more acute, not less, for early tokenized issues. Algorithmic compliance should support, never supplant, Shariah governance.

10 Limitations

The sukuk evidence rests on a single, highly illiquid vehicle; the illiquidity correction is informative but a constituent-level and multi-vehicle re-examination is needed before the spanning result is treated as settled. The data are predominantly United States listed and single currency, so Gulf-market and local-currency extensions are required. The sustainability analysis uses a green-bond proxy rather than green-sukuk-specific series and finds no measurable green premium, so the growing green-sukuk segment (Al Madani, Alotaibi, & Alhammadi, 2020) is left to dedicated data. Expense ratios used in the cost-of-opacity calculation are published figures that should be reconfirmed. The maqasid diagnostic is an interpretive frame, not a structural model, and the equity-null portion of the argument was specified before, while the framework itself was sharpened after, the first results, a sequence disclosed in the interest of candour.

11 Conclusion

Measured carefully, neither Shariah equity nor sukuk offers exposure that conventional risk factors do not already price. Screening relabels factors; the apparent distinctiveness of sukuk is, on the available evidence, the signature of a thinly traded market, though a single vehicle cannot separate illiquidity from the slow repricing of real assets. The most durable contribution is therefore methodological: a diagnostic that distinguishes the channels of compliance, and a caution that distinctiveness claimed for Islamic asset classes may be an artifact of testing illiquid instruments. For Islamic economics the implication is constructive rather than merely deflationary, in that the case for compliance need not rest on a differentiated return, and the agenda the evidence most supports is the deepening of Islamic capital markets, offered as a conjecture to be tested on Gulf-market data.

A Pre-registration

The estimands, identifying comparisons, and directional predictions were fixed before estimation. The matched ex-financials benchmark, the out-of-sample split (factors trained through 2019, evaluated 2020–2026), the tail and systemic predictions, and the illiquidity battery design (the frequency ladder and Dimson correction) were specified in advance, and multiple-testing was

anticipated. The maqasid diagnostic was sharpened after the equity null was observed; this sequence is disclosed in the interest of candour, and only the equity-null portion claims strict pre-specification.

Data and code availability

All data are free and public (full instrument list in Section 4; construction and parameters in Section 5). The complete analysis, from the data pull through every table and figure, is reproducible from a single command (`python reproduce.py`) under a permissive (MIT) licence. Code and processed outputs are available at [\[REPOSITORYURL\]](#) (archived at [\[ZENODO/OSFDOI\]](#)).

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